

Day 3 – Student Performance Prediction

Date: 2026-09-18

Topic: Machine Learning Fundamentals

Project: Student Performance Prediction (Regression + Classification)

What I learned

- What machine learning is and how the overall ML workflow fits together: dataset → features + target → train/test split → preprocessing → model training → prediction → evaluation.
- The difference between features (inputs like study time, absences, G1, G2) and the target/label (the value we want to predict, G3).
- Why we split data into training and testing sets — to evaluate the model on unseen data, the same way a student is tested on questions they haven't seen before.
- Regression predicts a numerical value (final grade). Classification predicts a category (Pass/Fail).
- Why different tasks need different metrics:
 - Regression → MAE, MSE, RMSE, R^2
 - Classification → Accuracy, Precision, Recall, F1-score, Confusion Matrix
- The difference between overfitting (model memorises training data) and underfitting (model is too simple to learn patterns).
- How preprocessing handles mixed data: StandardScaler for numerical features, One-Hot Encoding for categorical features.
- What each classification metric actually means:
 - Accuracy – overall percentage of correct predictions
 - Precision – of cases predicted positive, how many were actually positive
 - Recall – of actual positives, how many the model found
 - F1-score – balance between precision and recall
- How to read a confusion matrix (True Negatives, False Positives, False Negatives, True Positives).

What I implemented

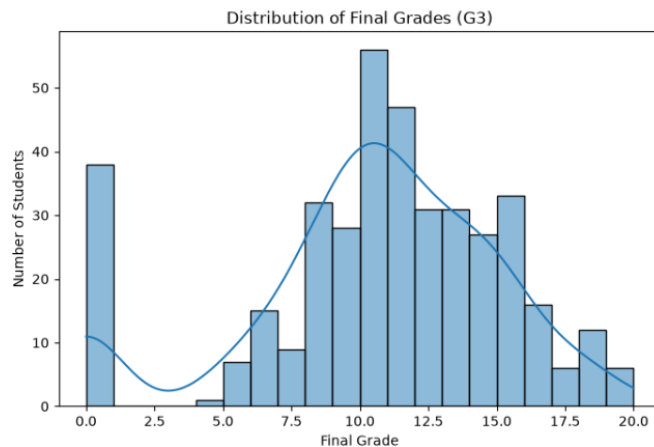
1. Loaded the UCI Student Performance dataset (student-mat.csv, semicolon-separated) using pandas.
2. Explored the dataset — shape, columns, data types, statistical summary.
3. Verified data quality: 0 missing values, 0 duplicate rows.
4. Visualised the distribution of the target variable G3 (final grade, range 0–20).

Explore the target (G3)

```
[37]: print(df["G3"].describe())
```

```
plt.figure(figsize=(8, 5))
sns.histplot(df["G3"], bins=20, kde=True)
plt.title("Distribution of Final Grades (G3)")
plt.xlabel("Final Grade")
plt.ylabel("Number of Students")
plt.show()
```

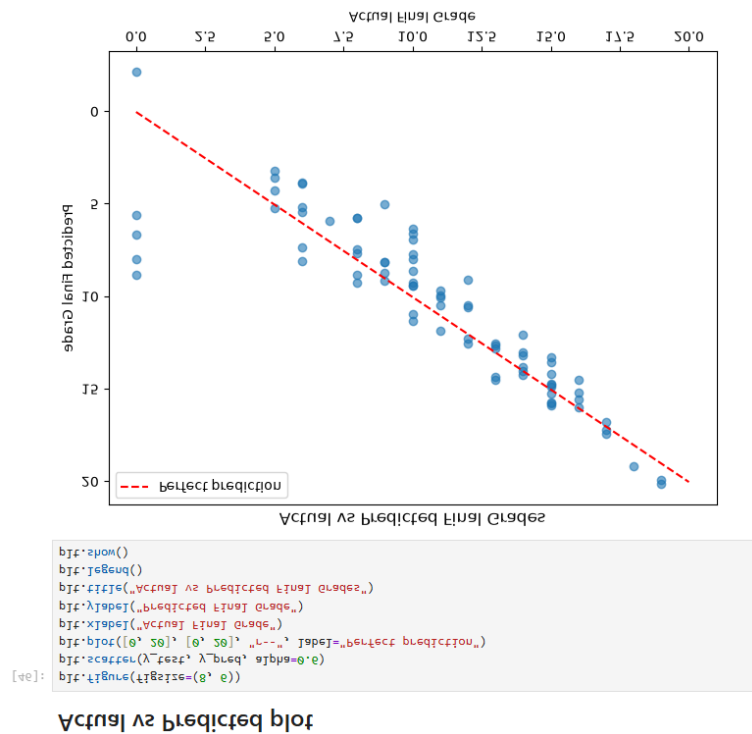
count	395.000000
mean	10.415190
std	4.581443
min	0.000000
25%	8.000000
50%	11.000000
75%	14.000000
max	20.000000
Name: G3, dtype: float64	



5. Selected 11 features (numerical + categorical) and set G3 as the target.
6. Split the data 80/20 into training and testing sets.
7. Built a preprocessing pipeline using ColumnTransformer with StandardScaler (numerical) and OneHotEncoder (categorical).
8. Built and trained a Linear Regression model to predict the final grade.
9. Evaluated the regression model with MAE, MSE, RMSE and R^2 .

10. Checked for overfitting by comparing training vs testing performance.

11. Plotted Actual vs Predicted final grades.



12. Created a binary Pass/Fail label from G3 (Pass = $G3 \geq 10$).

13. Performed a stratified train/test split for classification.

14. Built and trained a Logistic Regression model to predict Pass/Fail.

15. Evaluated the classification model with Accuracy, Precision, Recall and F1-score.

16. Generated a classification report and confusion matrix.

17. Tested both models on a new custom student and obtained a predicted grade and Pass/Fail result.

What worked

- Dataset loaded successfully using sep=";".
- Data preprocessing pipeline worked correctly for both numerical and categorical features.
- Both models (Linear Regression and Logistic Regression) trained without errors.

- Regression model produced accurate final-grade predictions.
- Classification model produced Pass/Fail predictions.
- All evaluation metrics were successfully calculated.
- Custom student prediction worked end-to-end for both models.

Final Results

Regression (Linear Regression) – predicting G3

```
MAE   : 1.4188
MSE   : 4.6417
RMSE  : 2.1545
R2   : 0.7736
```

```
Training MAE : 1.1569
Testing MAE   : 1.4188
Training R2  : 0.8443
Testing R2   : 0.7736
```

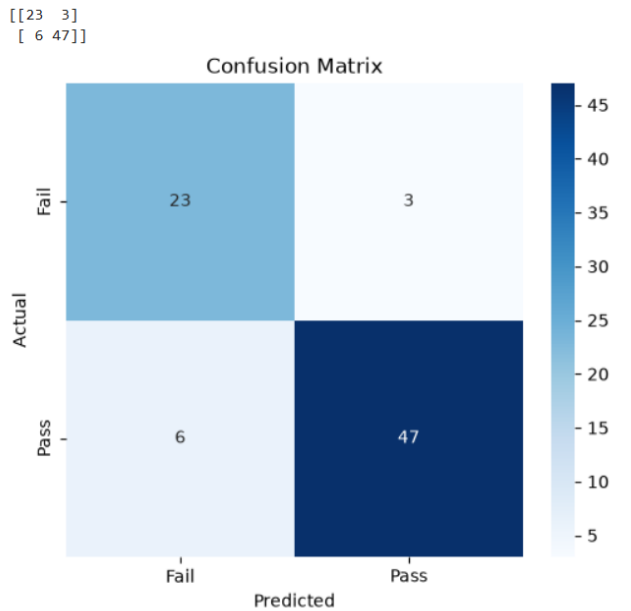
Interpretation: predictions are off by about 1.42 grade points on average, and the model explains about 77% of the variation in final grades. Training and testing scores are close, so there is no significant overfitting.

Classification (Logistic Regression) – predicting Pass/Fail

```
Accuracy : 0.8861
Precision: 0.9400
Recall   : 0.8868
F1-score : 0.9126
```

Confusion Matrix

	Predicted Fail	Predicted Pass
Actual Fail	23 (TN)	3 (FP)
Actual Pass	6 (FN)	47 (TP)



Custom student test

- Input: age 17, studytime 3, failures 0, absences 4, G1 = 15, G2 = 16
- Predicted final grade: 16.18
- Predicted status: PASS

What failed

1. At first, the dataset columns were not separated correctly when loading the CSV file.
2. Numerical features and categorical features should not be processed in exactly the same way.

How I solved it

1. I checked the dataset format and changed the Pandas loading code to:

```
df = pd.read_csv("data/student-mat.csv", sep=";")
```

```
[58]: df = pd.read_csv("data/student-mat.csv", sep=";")
      print("Shape:", df.shape)
      df.head()
```

Shape: (395, 33)

```
[58]:
```

	school	sex	age	address	famsize	Pstatus	Medu	Fedu
0	GP	F	18	U	GT3	A	4	4
1	GP	F	17	U	GT3	T	1	1
2	GP	F	15	U	LE3	T	1	1
3	GP	F	15	U	GT3	T	4	4
4	GP	F	16	U	GT3	T	3	3

2. I separated the features into numerical and categorical groups.

For numerical features, I used: StandardScaler

For categorical features, I used: OneHotEncoder

```
[0]: preprocessor = ColumnTransformer(
      transformers=[
          ("num", StandardScaler(), numerical_features),
          ("cat", OneHotEncoder(handle_unknown="ignore"), categorical_features)
      ]
    )

    print("Preprocessor created")
```

Preprocessor created

What I would improve

- Compare multiple algorithms (e.g. Decision Tree, Random Forest, Gradient Boosting) against Linear/Logistic Regression.
- Perform cross-validation for more reliable performance estimates.
- Tune model hyperparameters using GridSearchCV.
- Improve feature engineering — e.g. combine G1 and G2 into an average trend feature.
- Experiment with removing G1 and G2 from features (Version B) to test whether lifestyle and study factors alone can predict performance.

- Investigate the group of students with $G3 = 0$, since these are the hardest to predict and pull the R^2 down.